How Do We Quantify Sound Volume? Sound Intensity *Level* (Decibel)

SYNOPSIS

Thus far we have a pretty good idea of what sound *is*. This module turns to how we *experience* it — through hearing — and the scope is deliberately narrow: the emphasis is on loudness, on sound volume. What do we mean by sound volume, and how do we put a number on it? That second question is what this note answers, and the answer is the scale we will use for the rest of the module.

PERFORMANCE EXPECTATIONS

Recognize that the sound level is measured in decibels (dB). Understand that dB is a nonlinear scale. Understand that dB is a logarithmic scale. Quantify sound intensity from dB values.

Loudness and intensity are not the same word

Both words point at how high or low the volume is, but they are not interchangeable. **Loudness** is human perception — generally agreed human experience — so it can be subjective. **Sound intensity** is rigorously defined, and we can measure it precisely with instruments. From here on we will discuss sound intensity, and defer loudness until later.

Why a linear scale will not do

What loudness and intensity have in common is that neither one sits comfortably on a linear scale. We are used to counting bags of apples: one bag holds ten, so two bags hold twenty, and the numbers go up in even steps. That works because the range is small — a few to a hundred. Sound intensity does not behave that way. It goes up in **factors of 10**, and a quiet sound and a loud one can differ by a factor of a billion. You cannot handle numbers from the very small to the very large on one linear scale.

Here is the size of the problem, in your ears rather than in the arithmetic: to perceive a **doubling** of the loudness of one guitar, you need **ten guitars** playing the same thing at the same time.

The decibel

The sound level is measured on a **decibel** scale (dB). The dB scale measures the sound intensity I against a fixed **reference intensity** I_0, which is the human hearing threshold: $\mathbf{10^{-12}}$ **watts/meter**^{2}. Do not worry about the units yet. That is roughly a single grain of rice falling on a table. Breathing is thought to be about ten times this reference, so you can imagine how weak the reference is.

The scale works like this: the sound intensity increases **tenfold for every 10 dB**. By definition the reference has **0 dB** — which does not mean no sound, only that it is at zero *level*. Breathing is 10 dB, ten times the reference. Rustling leaves are 20 dB, ten times breathing and a hundred times the reference. An everyday conversation is 50 dB, which is 10^5 times the reference. Office noise is 60 dB, a million times the reference, and ten times an everyday conversation.

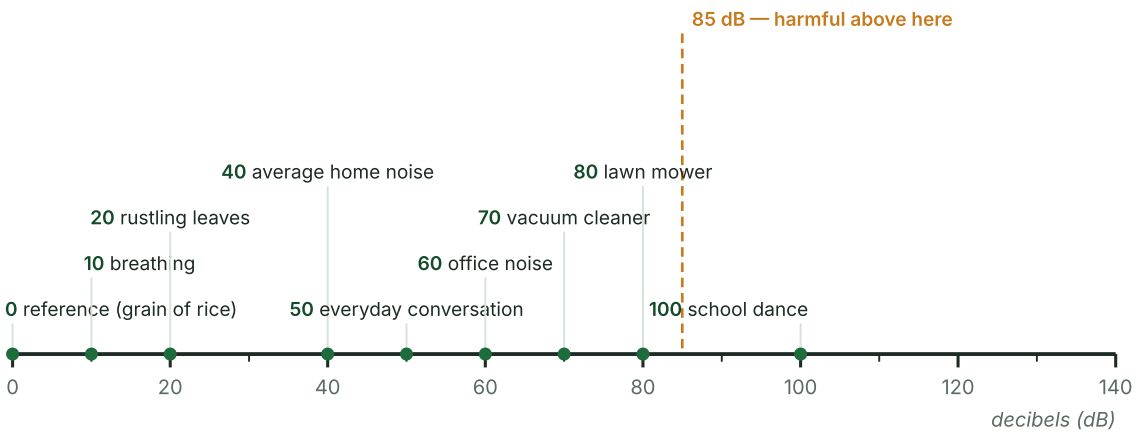


Figure 1. The decibel scale, with everyday sounds placed on it. Equal steps along the scale are equal *factors* in intensity, not equal amounts — that is what makes it nonlinear. Above 85 dB, sound is harmful to hearing.

Notice what that means at the top of the scale. At 100 dB, a school dance, you have increased the reference by **ten orders of magnitude** — ten billion-fold. When a quantity varies over a range like 1 to 10^{10}, we say it has a very large **dynamic range**. Our hearing sensitivity has a very large dynamic range.

TWO SHORTCUTS WORTH MEMORISING

Take the last zero off. What remains is the number of zeros in the factor you multiply the reference by. Take 80 dB, a lawn mower, close to the limit of being harmful: drop the one zero and you have 8, so it is a 10^8-fold increase — a hundred million times a grain of rice falling on a table. **And: a 3 dB increase is a doubling** of sound intensity. So if you have a 12 dB sound, a faint whisper, doubling that intensity gives you a 15 dB sound, a cat purring.



Figure 2. A difference in decibels against the factor the intensity changes by. The two shortcuts above are the first three columns; the last two follow from them.

In the following, we will take a deeper look at dB, the standard measure of sound intensity. Read the instructions and answer the questions.

Instructions and Questions

1. Because we are handling numbers going from very small (decimals with many zeros behind the decimal point) to very big (many zeros after a nonzero number), we will use a scale that essentially **counts the zeros**. The logarithmic scale is made to handle such a large dynamic range. On your calculator, the $\log_{10}$ button converts a number to a value on a log scale. The "10" in the subscript means we are using powers of 10. Fill in the blanks below.

Number	Same number in powers of ten	Log_{10} (number)
0.000001	10^{-6}	−6
0.00001	10^{-5}	−5
0.0001	10^{-4}	−4
0.001	10^{-3}	−3

Number	Same number in powers of ten	Log_{10} (number)
0.01	10^{-2}	−2
0.1	10^{-1}	−1
1	10^{0}	0
10	10^{1}	1
100	10^{2}	2
1000	10^{3}	3
10000	10^{4}	4
100000	10^{5}	5
1000000	10^{6}	6

2 When do you get a negative number for $\log_{10}$(number)?

> For **decimal numbers — any number less than 1**. 0.1, 0.01, 0.001 … give −1, −2, −3 … In other words 10^{x} where x is negative.

3 When do you get a positive number for $\log_{10}$(number)?

> For **numbers greater than 1**: 10, 100, 1000 … give 1, 2, 3 … In other words 10^{x} where x is positive.

4 What is the connection between the number and $\log_{10}$(number)?

> **$\log_{10}$(number) is the exponent the 10 is raised to** — the number of zeros.

5 The number 50 is between 10 and 100. What is $\log_{10}(10)$ and $\log_{10}(100)$?

If you took $\log_{10}(50)$, between which two numbers do you expect $\log_{10}(50)$ to be?

> $\log_{10}(10) = 1$ and $\log_{10}(100) = 2$, so $\log_{10}(50)$ lies **between 1 and 2**.
>
> But **not 1.5**. 50 sits halfway between 10 and 100 on a linear scale, and your instinct will say halfway between 1 and 2. In fact $\log_{10}(50) \approx$ **1.7**. That gap between 1.5 and 1.7 is the whole point of this note: **the scale is not linear**.

Put it in your calculator to confirm. Note that this number will not be right in the middle between $\log_{10}(10)$ and $\log_{10}(100)$.

6 We can get an exact number for $\log_{10}(50)$ if you write the math definition of log: $\log_{10}$(number) = (exponent with base 10). In other words, $\log_{10}(50) = 1.699$ means that $10^{1.699} = 50$. Confirm this by putting $10^{1.699}$ into your calculator. Do you get 50?

> Yes — $10^{1.699} = 50.0$ to three figures.
>
> The more formal way to write it: $\log_{10}x = y$ means $10^y = x$. For any number x we are looking for the exponent to raise 10 to so that we get x back. For 50 that exponent must be between 1 and 2, and it is 1.699.
>
> **log = logarithmic scale is non-linear.**

Now we are ready for dB. In the dB scale we take a log of the ratio I/I_0, where I is the sound intensity of interest and I_0 is the reference we are comparing it to. I_0 is about the sound intensity of a grain of rice falling on a table, 10^{-12} watts/meter^2. This is a fixed value. Then we multiply by 10.

$$\text{dB} = 10 \log_{10}(I / I_0) = 10 \log_{10}(I / 10^{-12})$$

7 So, for a quiet library at 40 dB, the actual sound intensity is 40 dB = $10 \log_{10}(I / 10^{-12})$. Dividing by 10 on both sides, $4 = \log_{10}(I / 10^{-12})$. You should be able to solve it by thinking about the number of zeros. Solve for I — it should have the unit of watts/meter^2.

$4 = \log_{10}(I / 10^{-12})$ means $I / 10^{-12} = 10^4$, so $\boldsymbol{I = 10^{-8}}$ **watts/meter^2**.

Said the way it is worth saying in class: ***I* is 10^4 times the reference sound — 10,000 times more energy**.

8

The sound intensity in absolute quantity (in watts/meter^2) is not as useful as the dB, because we only really want to get a feel for how loud a sound is. Comparing it to a known level is a better way to describe the sound volume. Here, the comparison is how many orders of magnitude — the number of zeros — the sound of interest is larger or smaller than the reference, because such a large scale of numbers is involved.

9

Answer the following questions:

i) How many times larger is the sound of a vacuum cleaner compared to the average home noise?

ii) You sing loudly at 90 dB. You want to sing even louder and raise the level to 93 dB. How much larger is your louder sound intensity?

iii) What is this intensity in part ii) in watts/meter^2?

iv) A sound below 55 dB is challenging to hear for individuals with moderate hearing impairment. If an everyday conversation is around 50 dB, how much larger does your sound intensity need to be for the hard-of-hearing person to hear you?

i) Vacuum cleaner 70 dB, average home noise 40 dB.
$70 = 10 \log_{10}(I_{vac} / 10^{-12}) \rightarrow 7 = \log_{10}(\ldots) \rightarrow I_{vac} = 10^{-5}$ W/m^2.
40 dB $\rightarrow I_{home} = 10^{-8}$ W/m^2. Ratio = 10^3 = **1000 times**.
The other route, worth doing out loud: 30 dB is ten steps of 3 dB, and every 3 dB doubles the intensity, so it is $2^{10} \approx 1000$. The two shortcuts agree.

ii) 93 − 90 = 3 dB, and 3 dB is a doubling, so **2 times**.

iii) 90 dB gives $I = 10^9 \times 10^{-12} = 10^{-3}$ W/m^2. 93 dB gives $9.3 = \log_{10}(I/10^{-12})$, so $I = 10^{9.3} \times$

$10^{-12} = 10^{-2.7} =$ **1.995×10^{-3} W/m^2** — about twice 1×10^{-3}, which is the check on part ii).

iv) $I_1 = 10^{5.5} \times 10^{-12}$ and $I_2 = 10^5 \times 10^{-12}$, so $I_1/I_2 = 10^{0.5} =$ **3.2 times** the intensity.

Files for this note

PRINT

Guided note (PDF)

Print it out and write on it in class.

READ

Note with answers shown (PDF)

The worked-through version, for review after class.

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